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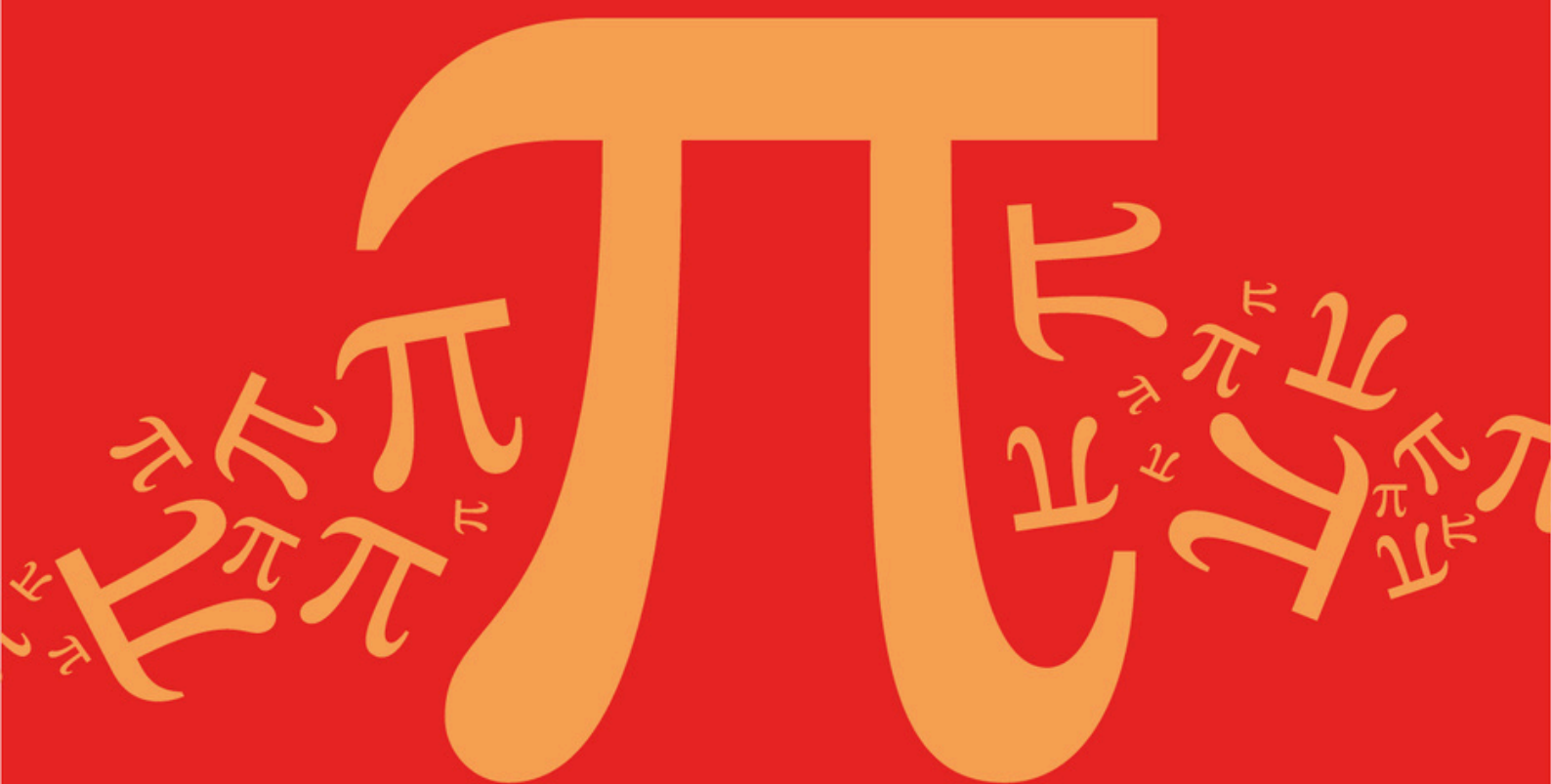
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AQA A Level Further Mathematics (7367)



Topic

C: Matrices

Sub topic

C6: Singular Matrices and Inverse of a 3x3 Matrix

Topic Questions

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Question Paper

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C6: Singular Matrices and Inverse of a 3x3 Matrix

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**To be used by all students and teachers for
AQA A Level Further Mathematics (7367)
for exams from 2025 onwards.**

Students and teachers of other boards may also find this useful

Name : _____

Date : _____

Time : _____

Total Marks Available : _____

Total Marks Received : _____

Q1.

Three planes have equations

$$x - y + kz = 3$$

$$kx - 3y + 5z = -1$$

$$x - 2y + 3z = -4$$

where k is a real constant. The planes do not meet at a unique point.

- (a) Find the possible values of k

(3)

- (b) There are two possible geometric configurations for the given planes.

Identify each possible configuration, stating the corresponding value of k

Fully justify your answer.

(5)

- (c) Given further that the equations of the planes form a consistent system, find the solution of the system of equations.

(3)

(Total 11 marks)

Q2.

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- (a) By direct expansion, or otherwise, show that the value of

$$\begin{vmatrix} -2 & 1 & 2k \\ -1 & 1 & k+1 \\ 2 & k-1 & 1 \end{vmatrix}$$

is independent of k .

(4)

- (b) (i) State, with a reason, whether the equations

$$-2x + y + 6z = 1$$

$$-x + y + 4z = 0$$

$$2x + 2y + z = -1$$

are consistent or inconsistent.

(2)

- (ii) The three equations given in part (b)(i) are the Cartesian equations of three planes.

State the geometrical configuration of these three planes.

(1)

(Total 7 marks)

Q3.

The system of equations

$$\begin{aligned} 2x - y - z &= 3 \\ x + 2y - 3z &= 4 \\ 2x + y + az &= b \end{aligned}$$

does not have a unique solution.

(a) Show that $a = -3$.

(3)

(b) Given further that the equations are inconsistent, find the possible values of b .

(2)

(Total 5 marks)

Q4.

The 3×3 matrices **A** and **B** satisfy

$$\mathbf{AB} = \begin{bmatrix} k & 8 & 1 \\ 1 & 1 & 0 \\ 1 & 4 & 0 \end{bmatrix}, \text{ where } \mathbf{A} = \begin{bmatrix} k & 6 & 8 \\ 0 & 1 & 2 \\ -3 & 4 & 8 \end{bmatrix}$$

and k is a constant.

(a) Show that **AB** is non-singular.

(1)

(b) Find $(\mathbf{AB})^{-1}$ in terms of k .

(5)

(c) Find $\mathbf{B}^{-1}$.

(4)

(Total 10 marks)

Q5.

(a) Determine the two values of the integer n for which the system of equations

$$2x + ny + z = 5$$

$$3x - y + nz = 1$$

$$-x + 7y + z = n$$

does not have a unique solution.

(4)

(b) For the positive value of n found in part (a), determine whether the system is consistent or inconsistent, and interpret this result geometrically.

(6)

(Total 10 marks)

Q6.

The matrix $\mathbf{A} = \begin{bmatrix} k & 1 & 2 \\ 2 & k & 1 \\ 1 & 2 & k \end{bmatrix}$, where k is a real constant.

- (a) (i) Show that there is a value of k for which

$$\mathbf{A} \mathbf{A}^T = m \mathbf{I}$$

where m is a rational number to be determined and $\mathbf{I}$ is the 3×3 identity matrix.

(6)

- (ii) Deduce the inverse matrix, $\mathbf{A}^{-1}$, of $\mathbf{A}$ for this value of k .

(1)

- (b) (i) Find $\det \mathbf{A}$ in terms of k .

(2)

- (ii) In the case when $\mathbf{A}$ is singular, find the integer value of k and show that there are no other possible real values of k .

(3)

- (iii) Find the value of k for which $\lambda = 7$ is a real eigenvalue of $\mathbf{A}$.

(2)

(Total 14 marks)

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Q7.

$$\mathbf{X} = \begin{bmatrix} 3 & 1 & -3 \\ 2 & 4 & 3 \\ -4 & 2 & -1 \end{bmatrix}$$

The non-singular matrix

- (a) (i) Show that $\mathbf{X}^2 - \mathbf{X} = k\mathbf{I}$ for some integer k .

(3)

- (ii) Hence show that $\mathbf{X}^{-1} = \frac{1}{20}(\mathbf{X} - \mathbf{I})$.

(2)

$$\mathbf{Y}^{-1} = \begin{bmatrix} 60 & 0 & 0 \\ 0 & 0 & -10 \\ 0 & 20 & 0 \end{bmatrix}$$

- (b) The 3×3 matrix $\mathbf{Y}$ has inverse

Without finding $\mathbf{Y}$, determine the matrix $(\mathbf{XY})^{-1}$.

(3)

(Total 8 marks)

Q8.

The matrices **A** and **B** are defined in terms of a real parameter t by

$$\mathbf{A} = \begin{bmatrix} 1 & 2 & 1 \\ 2 & t & 4 \\ 3 & 2 & -1 \end{bmatrix} \text{ and } \mathbf{B} = \begin{bmatrix} 15 & -4 & -1 \\ -2t & 4 & 2 \\ 17 & -4 & -3 \end{bmatrix}$$

- (a) Find, in terms of t , the matrix **AB** and deduce that there exists a value of t such that **AB** is a scalar multiple of the 3×3 identity matrix **I**.

(5)

- (b) For this value of t , deduce $\mathbf{A}^{-1}$.

(2)

(Total 7 marks)

Q9.

The matrix $\mathbf{A} = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 3 \\ 4 & 3 & k \end{bmatrix}$, where k is a constant.

Determine, in terms of k where appropriate:

- (a) $\det \mathbf{A}$;

(2)

- (b) $\mathbf{A}^{-1}$.

(5)

(Total 7 marks)

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Q10.

The matrix $\mathbf{P} = \begin{bmatrix} 4 & -1 & 2 \\ 1 & 1 & 3 \\ -2 & 0 & a \end{bmatrix}$, where a is constant.

- (a) (i) Determine $\det \mathbf{P}$ as a linear expression in a .

(2)

- (ii) Evaluate $\det \mathbf{P}$ in the case when $a = 3$.

(1)

- (iii) Find the value of a for which **P** is singular.

(2)

- (b) The 3×3 matrix **Q** is such that $\mathbf{PQ} = 25\mathbf{I}$.

Without finding **Q**:

- (i) write down an expression for $\mathbf{P}^{-1}$ in terms of **Q**;

- (1)
- (ii) find the value of the constant k such that $(\mathbf{PQ})^{-1} = k\mathbf{I}$; (2)
- (iii) determine the numerical value of $\det \mathbf{Q}$ in the case when $a = 3$. (4)
- (Total 12 marks)

Q11.

The matrices $\mathbf{P}$ and $\mathbf{Q}$ are given by

$$\mathbf{P} = \begin{bmatrix} 2 & 1 & 1 \\ 1 & t & -2 \\ 3 & 2 & 1 \end{bmatrix} \quad \text{and} \quad \mathbf{Q} = \begin{bmatrix} 1 & 1 & 1 \\ -7 & -1 & 5 \\ 11 & -1 & -7 \end{bmatrix}$$

where t is a real constant.

- (a) Find the value of t for which $\mathbf{P}$ is singular. (2)
- (b) (i) Determine the matrix $\mathbf{R} = \mathbf{PQ}$, giving its elements in terms of t where appropriate. (3)
- (ii) Find the value of t for which $\mathbf{R} = k\mathbf{1}$, for some integer k . (2)
- (iii) Hence find the matrix $\mathbf{Q}^{-1}$. (1)
- (c) In the case when $t = -3$, describe the geometrical transformation with matrix $\mathbf{R}$. (2)
- (Total 10 marks)